

NLP and classification [Theory MCQ] (Version : 0)

TEST

● **Correct Answer**

🕒 Answered in 2.72 Minutes

Question 1/15

Which of the following research scenarios could be solved using a classification model?

☐

Fitting a generative model to a set of cat images and then generating a new cat image.

☐

Predicting the time of rehabilitation for COVID-19 infected patients.



Sorting aerial photographs into urban and agricultural zones.

☐

Predicting the number of litres of beer consumed after level 3 lockdown restrictions have been implemented.

Explanation:

“Predicting the time of rehabilitation for COVID-19 infected patients” is incorrect because it is a regression problem, not a classification problem.

“Fitting a generative model to a set of cat images and then generating a new cat image” is incorrect because it involves generative modelling, not classification.

“Predicting the number of litres of beer consumed after level 3 lockdown restrictions have been implemented” is incorrect because it is a regression problem, not a classification problem.

Question 2/15

Which statement regarding the assessment of a classification model is true?

☒ The ROC curve can display both true positive and false positive rates at a range of thresholds.

☐ Recall refers to the percentage of total incorrect classifications by a model.

☐ AUC graphs do not show the overall performance of a classifier.

☐ Precision is calculated as the number of true positives divided by the sum of true positives and true negatives.

Explanation:

“AUC graphs do not show the overall performance of a classifier” is incorrect because AUC (Area Under the Curve) is commonly used to evaluate the overall performance of a classifier.

“Recall refers to the percentage of total incorrect classifications by a model” is incorrect because recall measures the proportion of actual positives that were correctly identified.

“Precision is calculated as the number of true positives divided by the sum of true positives and true negatives” is incorrect because precision measures the proportion of true positives among all positive predictions made by the model.

Question 3/15

What will the following piece of code output?

```
from sklearn.metrics import classification_report
print('Classification Report')
print(classification_report(y_test, pred_lm, target_names=['ham', 'spam']))
```

☒ A matrix for assessing precision, recall, f1-score and support for classifying "ham" or

"spam", along with micro, macro and weighted averages for the model.

☐ A list of all the correct and incorrect classifications for "ham" or "spam" predicted by the model.

☐ A 2x2 confusion matrix for classifying "ham" or "spam".

☐ A table of accuracies for "spam" and "ham".

Explanation:

'A table of accuracies for "spam" and "ham"'. This is incorrect because the `classification_report` function doesn't output a simple table of accuracies but rather a detailed report including precision, recall, f1-score, and support for each class.

'A 2x2 confusion matrix for classifying "ham" or "spam"'. This is incorrect because the `classification_report` function does not output a confusion matrix.

'A list of all the correct and incorrect classifications for "ham" or "spam" predicted by the model.' This is incorrect because the `classification_report` function does not provide a list of individual correct and incorrect classifications.

Question 4/15

Which of the following statements is correct?

- i) Sensitivity refers to the True Positive Rate
- ii) F1-Score is the harmonic mean between precision and recall
- iii) "LogisticRegression" can be imported from "sklearn.linear_model"
- iv) SVM and Random Forests can be used for classification

☐ ii), iii)

☐ i), iii)

☒ All of the above

☐ i), ii), iii)

Explanation:

ii), iii) - This is incorrect because it only includes two correct statements, missing out on the correct statement regarding Sensitivity (i).

i), ii), iii) - This is incorrect because it does not include the correct statement regarding SVM and Random Forests for classification (iv).

i), iii) - This is incorrect because it misses out on the correct statement about F1-Score (ii) and SVM and Random Forests for classification (iv).

Question 5/15

Which one of the following statements is correct regarding the parameter C in logistic regression?

☐ C is directly proportional to the regularisation coefficient, lambda.

☒ An increase in C will result in less regularisation.

☐ C can be assigned a number less than or equal to zero.

☐ C can only be used for binary logistic regression.

Explanation:

"C is directly proportional to the regularisation coefficient, lambda" is incorrect because C is inversely proportional to lambda, and an increase in C leads to less regularisation, not more.

"C can be assigned a number less than or equal to zero" is incorrect because C must be a positive

value.

“C can only be used for binary logistic regression” is incorrect because C is a parameter used in logistic regression for both binary and multiclass classification.

Question 6/15

You built a model which classifies if you're a python-wiz or a python-was. Calculate the F1-Score based on the following data:

Correctly identified as a python-wiz : 126

Correctly identified as a python-was : 58

Incorrectly identified as a python-wiz: 14

Incorrectly identified as a python-was: 22

☐ 0.85

☐ 0.84

☐ 0.89

☒ 0.88

Explanation:

0.85 is incorrect because it's not the correct F1-Score calculated based on the provided data. 0.89 is incorrect because it's not the correct F1-Score calculated based on the provided data. 0.84 is incorrect because it's not the correct F1-Score calculated based on the provided data.

Question 7/15

You built a model to detect COVID-19 cases in rural areas based on daily population movements. Your classifier had the following results:

Correctly identified negative cases: 100241

Correctly identified positive cases: 95

Incorrectly identified negative cases: 24020

Incorrectly identified positive cases: 28

Which of the following metrics would best assess our model's performance?

☒ F1-Score

☒ Recall

☐ Accuracy

☐ Precision

Explanation:

Accuracy - This is incorrect because accuracy doesn't consider class imbalance and it might not be the best metric in this case, where the classes are imbalanced.

Precision - This is incorrect because precision focuses on the proportion of true positive predictions among all positive predictions and may not be suitable when false negatives are important, like in the case of COVID-19 detection.

Recall - This is incorrect because recall focuses on the proportion of true positive predictions among all actual positive instances and may not be sufficient alone to evaluate the model's performance.

Question 8/15

When building a logistic regression model, we can do the following:

☐ Change the value of C to -1.00

☒ Change the probability threshold from 0.5 to 0.6

☐ Automatically optimise the C parameter in the LogisticRegression() object

☐ Apply dropout regularisation to the model

Explanation:

Change the value of C to -1.00 - This is incorrect because the parameter C in logistic regression must be a positive value, representing the inverse of regularisation strength. It cannot be negative.

Apply dropout regularisation to the model -This option is incorrect because dropout regularisation is predominantly applied in neural networks to counter overfitting. In contrast, logistic regression models commonly employ methods such as L1 or L2 regularisation to mitigate overfitting, making dropout regularisation unsuitable for logistic regression.

Automatically optimise the C parameter in the LogisticRegression() object - This is incorrect because while some algorithms allow for automatic optimization of parameters, it's not a standard practice in logistic regression, where you usually need to specify the value of C manually.

Question 9/15

Assume we have done the following:

- Imported LogisticRegression from sklearn.linear_model
- Imported train_test_split from sklearn.model_selection
- Cleaned and split your dataset into X and y

Select the lines of code that have an error.

- i) `X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=42)`
- ii) `model = LogisticRegression(C=0.1, penalty='l1')`
- iii) `model.fit(X_train_y_train)`
- iv) `pred = model.predict(y_test)`

☐ i and iv

☒ iii and iv

☐ ii

☐ iv only

Explanation:

iv only - This is incorrect because not only does the

prediction line (iv) have an error (it should be `pred = model.predict(X_test)`) but the fitting line (iii) also has an error as it should fit the model using both `X_train` and `y_train`, not `X_train_y_train`.

i and iv - This is incorrect because only the fitting line (iv) has an error. The splitting line (i) is correct.

ii - This is incorrect because the model initialization line (ii) is correct.

Question 10/15

Which of the following statements is CORRECT?

☐

We can get more accurate predictions on a binary classification dataset by training a logistic regression model on each class individually and then adding the two model predictions together.

☐

The more data we have, the faster our logistic regression model trains.

☐

The logistic regression model can detect and remove outliers in the data.



Logistic Regression is a parametric model whereas a K-Nearest Neighbours model is non-parametric.

Explanation:

“We can get more accurate predictions on a binary classification dataset by training a logistic regression model on each class individually and then adding the two model predictions together” is incorrect because it's not a technique used for improving classification accuracy.

“The more data we have, the faster our logistic regression model trains” is incorrect because the training time of a logistic regression model is not directly proportional to the amount of data.

“The logistic regression model can detect and remove outliers in the data” is incorrect because logistic regression itself does not inherently detect

or remove outliers.

Question 11/15

What is the purpose of the softmax activation function in the output layer of a neural network?



To normalise the output into a probability distribution.



To calculate the gradient during backpropagation.



To control the convergence of the optimisation algorithm.



To introduce non-linearity into the model.

Explanation:

“To introduce non-linearity into the model” is incorrect because softmax is primarily used for probability normalisation, not introducing non-linearity.

“To calculate the gradient during backpropagation” is incorrect because softmax is not directly involved in calculating gradients during backpropagation.

“To control the convergence of the optimization algorithm” is incorrect because the softmax activation function is used for normalising the output into a probability distribution. It does not control the convergence of the optimisation algorithm.

Question 12/15

What is the purpose of the ReLU activation function in a neural network?



To introduce non-linearity into the model.



To normalise the output into a probability distribution.

☐ To introduce linearity into the model.

☐ To calculate the gradient during backpropagation.

Explanation:

“To introduce linearity into the model.” - This option is Incorrect because ReLU introduces non-linearity, not linearity.

“To normalise the output into a probability distribution.” - This option is Incorrect because ReLU is not used for normalisation; it's primarily used for introducing non-linearity.

“To calculate the gradient during backpropagation.” - This option is Incorrect because ReLU is used for activation, not for calculating gradients during backpropagation.

Question 13/15

In the following code snippet, what is the role of the `validation_split` argument in the `.fit()` method?

Compile model

```
model.compile(  
    optimizer='adam',  
    loss='categorical_crossentropy',  
    metrics=['accuracy'],  
)
```

Training the model

```
history = model.fit(x_train, y_train, batch_size=32, epochs=10, verbose=True, validation_split=0.1)
```

☐ It specifies the number of epochs for training.

☐ It controls the learning rate during training.

☐ It splits the dataset into training and

testing sets.



It specifies the fraction of the training data to be used as validation data.

Explanation:

“It splits the dataset into training and testing sets.”

This option is incorrect because the `validation_split` argument only splits the training data for validation, not into training and testing sets.

“It specifies the number of epochs for training.” -

This option is incorrect because the number of epochs is specified separately, not with the `validation_split` argument.

“It controls the learning rate during training.” - This option is incorrect because the learning rate is controlled by the optimiser chosen during model compilation, not by the `validation_split` argument.

Question 14/15

Which of the following TensorFlow/Keras callbacks is commonly used to stop training when a monitored metric has stopped improving?



ReduceLROnPlateau



TensorBoard



EarlyStopping



ModelCheckpoint

Explanation:

ModelCheckpoint is incorrect because it is used to save the model's weights during training.

ReduceLROnPlateau is incorrect because it is used to reduce the learning rate when a monitored metric has stopped improving.

TensorBoard is incorrect because it is used for visualisation and monitoring training progress, but not for stopping training based on a metric.

Question 15/15

Which one of the following code snippets correctly compiles a TensorFlow/Keras model for binary classification with binary cross-entropy loss and accuracy metric?

☐ `model.compile(optimizer='adam',
loss='categorical_crossentropy', metrics=
['accuracy'])`

☒ `model.compile(optimizer='rmsprop',
loss='binary_crossentropy', metrics=
['accuracy'])`

☐ `model.compile(optimizer='sgd',
loss='binary_crossentropy', metrics=
[mean_squared_error])`

☐ `model.compile(optimizer='adam',
loss='mean_squared_error', metrics=
['accuracy'])`

Explanation:

“`model.compile(optimizer='adam',
loss='categorical_crossentropy', metrics=
['accuracy'])`” is incorrect because it uses categorical cross-entropy loss which is suitable for multi-class classification, not binary classification.

“`model.compile(optimizer='sgd',
loss='binary_crossentropy', metrics=
[mean_squared_error])`” is incorrect because it specifies “mean_squared_error” as a metric which is not appropriate for binary classification tasks but for regression problems.

“`model.compile(optimizer='adam',
loss='mean_squared_error', metrics=['accuracy'])`” is incorrect because it uses mean squared error loss which is not suitable for classification tasks.

